

Pre Lab 5

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Q.1) Using Fig 5.1 with $V_{in} = 3V$ sine at $10kHz$, $R = 1k\Omega$ and $C = 10nF$, determine the theoretical capacitive reactance (X_c) and circuit impedance ($Z < \theta$) and record results in Table 5.2. Also calculate the theoretical values of α and fill Table 5.1.

$$X_c = \frac{-j}{2\pi fC} = \frac{-j}{2\pi(10k)(10nF)} = 1591.55 \Omega$$

* $-j$ only represents that it's the imaginary part which is the reactive component.

$Z < \theta$

$$|Z| = \sqrt{(1k)^2 + (1591.55)^2} = 1879.64 \Omega$$

$$\theta = \tan^{-1}\left(\frac{1591.55}{1000}\right) = +57.87^\circ$$

Frequency	$X_c(Th.)$	$Z < \theta(Th.)$			For Resistor	For Capacitor
10K	1591.5 Ω	1879.64 / 57.87°				
20K	795.77 Ω	1277.98 / 38.5°		α Theoretical	0°	90°
50K	318.3 Ω	1049.4 / 17.66°		Leading component	-	Current
100K	159.15 Ω	1012.59 / 9.04°				

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Q.2) Using Fig 5.2 with $V_{in} = 3V$ sine at $10kHz$, $R = 1k\Omega$ and $L = 10mH$, determine the theoretical inductive reactance (X_L) and circuit impedance ($Z \angle \theta$) and record the results in Table 5.4 and also calculate theoretical value of β and fill Table 5.4.

$$\underline{X_L} = j 2\pi f L \Rightarrow 2\pi (10k)(10m) \Rightarrow 628.32 \Omega$$

$$\underline{|Z|} = \sqrt{(1k)^2 + (628.32)^2} \Rightarrow 1184.03 \Omega$$

$$\underline{\theta} = \tan^{-1} \left(\frac{X_L}{R} \right) = \tan^{-1} \left(\frac{628.32}{1000} \right) = 32.14^\circ$$

Frequency	X_L (Th.)	$Z \angle \theta$ (Th.)	β Theoretical	Resistor	Inductor
			Leading component	0°	90°
10k	628.32 Ω	1184.03 $\angle 32.14^\circ$		—	Voltage
20k	1256.63 Ω	1605.96 $\angle 51.48^\circ$			
50k	3141.6 Ω	3296.9 $\angle 72.34^\circ$			
100k	6283.2 Ω	6362.3 $\angle 80.96^\circ$			

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Q.3) Using Fig 5.3 with $V_{in} = 3V$ sine at $10kHz$, $R = 1k\Omega$ and $C = 10nF$ determine the theoretical capacitive reactance (X_c) and circuit impedance (Z $\angle \theta$) and record results in Table 5.5.

$$\underline{X_c} = \frac{1}{2\pi f C} = \frac{1}{2\pi(10k)(10n)} = 1591.55 \Omega$$

$$\underline{Z} = \frac{R X_c^2}{R^2 + X_c^2} - j \frac{R^2 X_c}{R^2 + X_c^2}$$

$$= \frac{(1000)(1591.55)^2}{(1000)^2 + (1591.55)^2} - j \frac{(1000)^2 (1591.55)}{(1000)^2 + (1591.55)^2}$$

$$Z = 716.96 - j450.5$$

 $\approx$
 ~~1591.55×10^5~~

Frequency	X_c (Th.)	Z (Th.)
5k	3183.1 Ω	28 910.16 - j285.9
20k	795.77 Ω	387.72 - j487.23
50k	318.3 Ω	91.99 - j289.02
100k	159.15 Ω	24.7 - j155.22